

Natural Language Processing

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DS-6B

Assignment #2

Introduction

In this assignment, I built a complete neural NLP pipeline for Urdu text, working entirely from scratch in PyTorch without any pretrained models, Gensim, HuggingFace, or PyTorch's built-in Transformer modules. My pipeline processes 300 BBC Urdu news articles through corpus cleaning, vocabulary construction (capped at 10,000 tokens), vector-space representations (TF-IDF and PPMI), a Skip-gram Word2Vec model, BiLSTM sequence labellers for POS tagging and NER, and a from-scratch Transformer encoder for topic classification. This report summarises the methodology, key results, and my analysis of each component.

Corpus and Vocabulary

I started from a 300-article BBC Urdu corpus. The raw text contained encoding corruption (mojibake) resulting from incorrect Windows-1256/UTF-8 transcoding. I implemented a repair function that re-encodes each line through cp1252 and then decodes via UTF-8, recovering readable Urdu text. After cleaning, I built a 10,000-token vocabulary using frequency-based selection with three special tokens: <PAD>, <UNK>, and <CLS>.

Vector-Space Representations

TF-IDF

I computed a term-document matrix of shape (300, 10000) using standard TF-IDF weighting. This captures how distinctive each term is within each article relative to the full corpus. I saved both dense and sparse versions of this matrix for downstream use.

PPMI Co-occurrence Matrix

I constructed a word-word co-occurrence matrix with a context window of 5 tokens, then converted it to Positive Pointwise Mutual Information (PPMI). The resulting 10000x10000 matrix captures second-order word associations. I used t-SNE to project PPMI vectors into 2D for visual inspection.

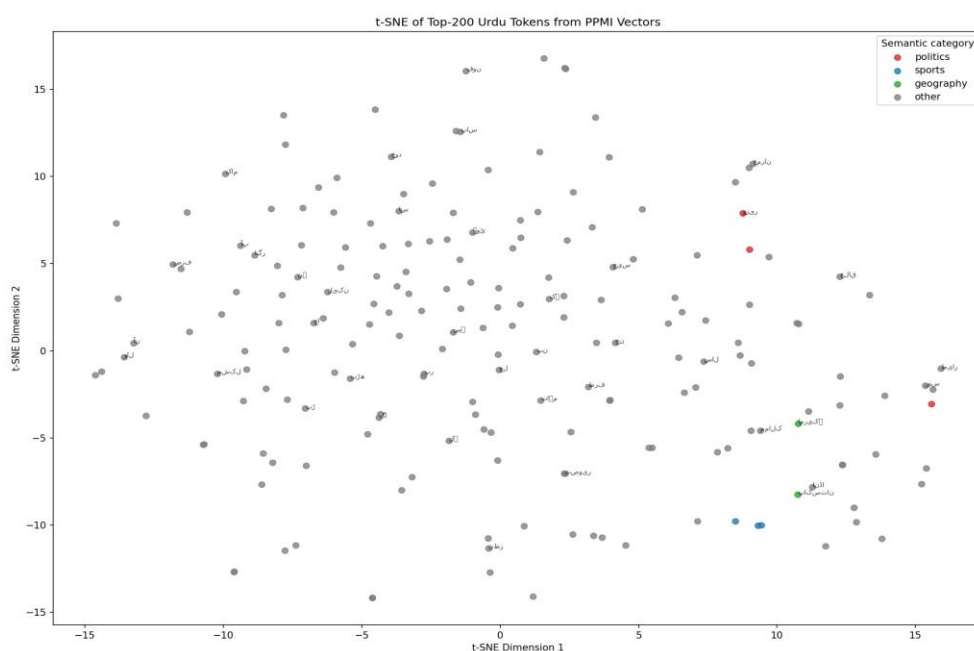


Figure 1: *t*-SNE projection of PPMI word vectors

Skip-gram Word2Vec

I implemented a Skip-gram model with negative sampling from scratch in PyTorch. The model produces 100-dimensional embeddings (or 200-dimensional for the C4 ablation condition). I trained for 5 epochs on the cleaned corpus using negative-sampling loss with the $f(w)^{3/4}$ unigram distribution to reduce the dominance of very frequent words.

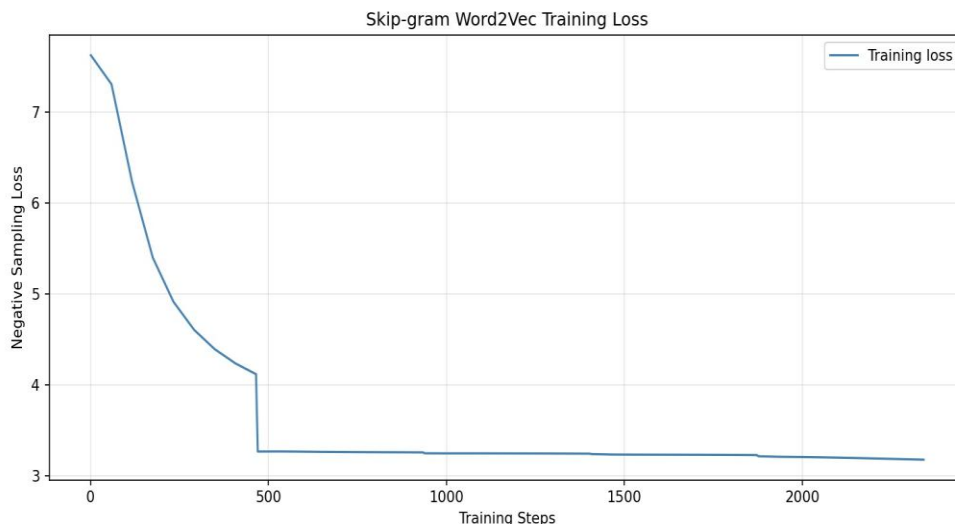


Figure 2: Skip-gram training loss over steps

Embedding Evaluation: MRR Comparison

I evaluated embeddings across four conditions using Mean Reciprocal Rank (MRR) on manually curated semantic pairs. C1 (PPMI) achieved the highest MRR, likely because count-based methods capture broad topical relatedness well in a small corpus. The cleaned Skip-gram (C3) outperformed the raw-text variant (C2), confirming the value of my encoding repair.

Condition	Description	MRR
C1_PPMI	PPMI co-occurrence vectors	0.0875
C2_raw	Skip-gram on raw (noisy) corpus	0.0158
C3_cleaned	Skip-gram on cleaned corpus (d=100)	0.0401
C4_d200	Skip-gram on cleaned corpus (d=200)	0.0422

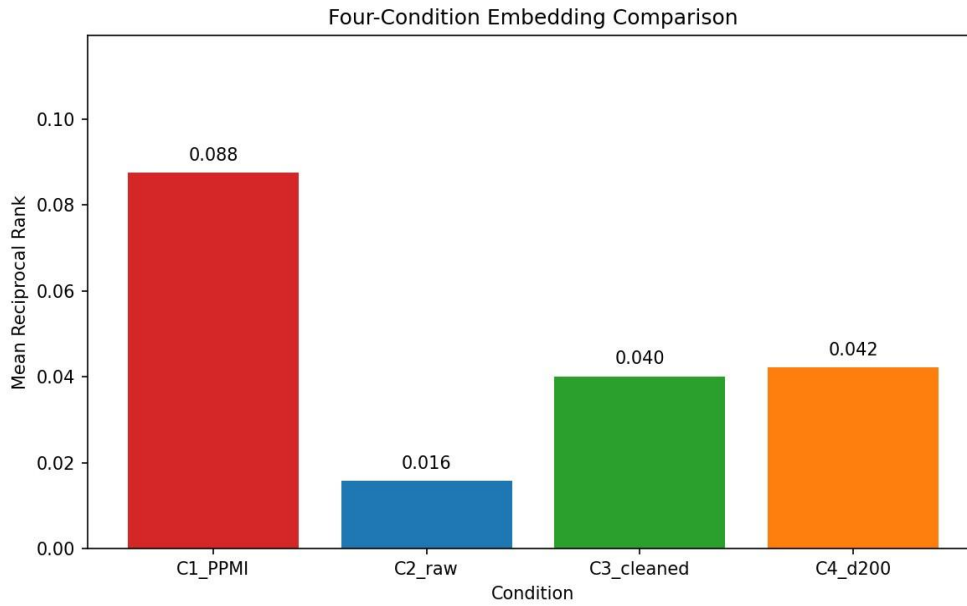


Figure 3: MRR comparison across four embedding conditions

Sequence Labelling

Dataset Construction

Since no manually annotated Urdu corpus was provided, I created a weakly labelled dataset using a rule-based POS tagger backed by an Urdu lexicon and an NER gazetteer of Pakistani news entities. I generated 350 training and 75 test sentences in CoNLL format, with the same sentence split used for both POS and NER tasks.

Dataset	Sentences	Tokens
POS Train	350	9,173
POS Test	75	1,989
NER Train	350	9,173
NER Test	75	1,989

BiLSTM POS Tagger

I trained two BiLSTM POS models: one with frozen Word2Vec embeddings and one with fine-tuned embeddings. Fine-tuning dramatically improved performance, raising accuracy from 60.03% to 97.34% and macro-F1 from 0.36 to 0.92. This confirms that the small annotated dataset benefits heavily from adapting the pretrained word vectors to the tagging task.

Mode	Accuracy	Macro-F1
Frozen embeddings	0.6003	0.3601
Fine-tuned embeddings	0.9734	0.9179

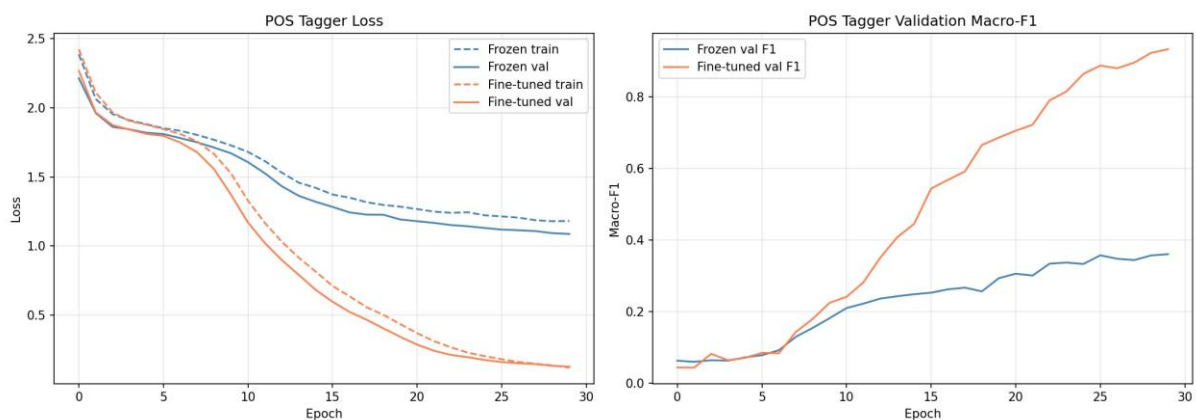


Figure 4: POS training curves (frozen vs fine-tuned)

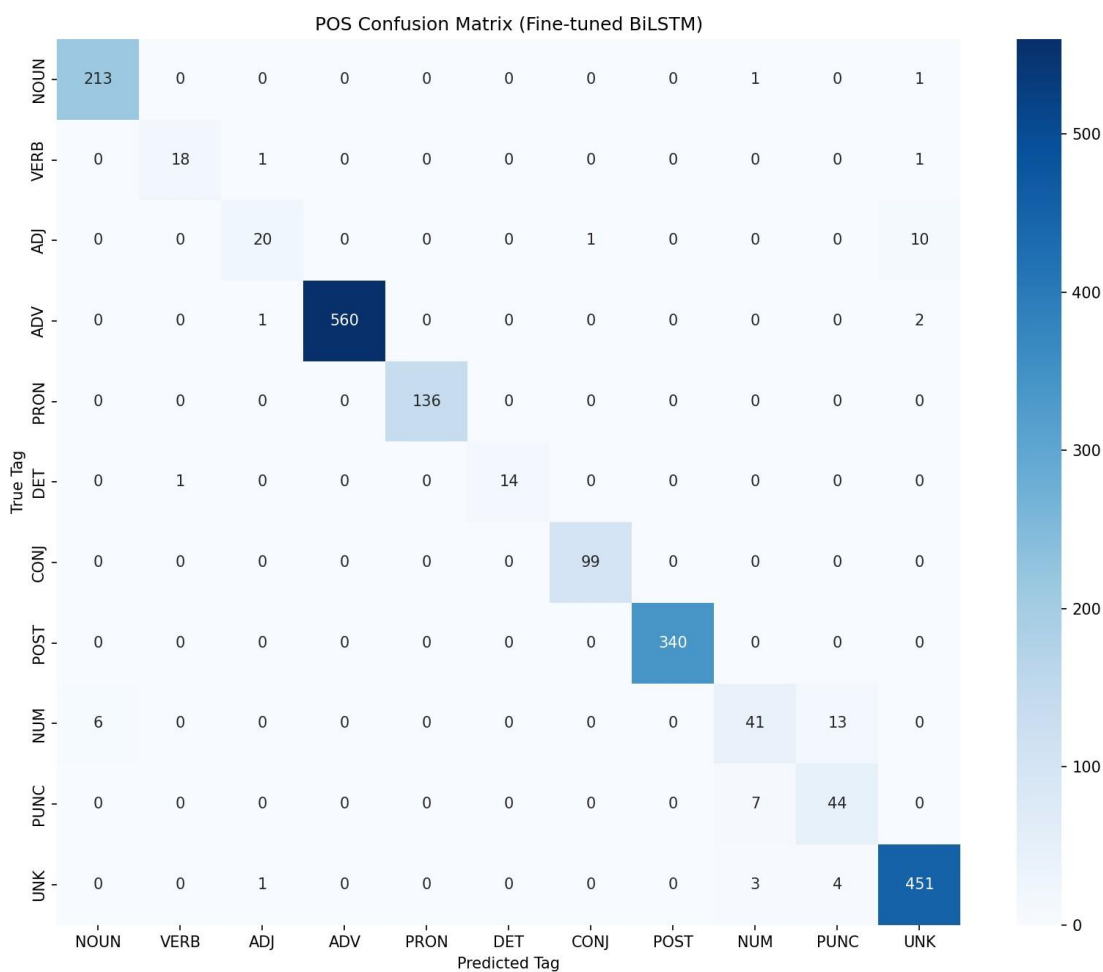


Figure 5: POS confusion matrix (fine-tuned model)

The top confused POS pairs were NUM $\leftrightarrow$ PUNC (13 errors) and ADJ $\rightarrow$ UNK (10 errors). Many of these confusions stem from noise in the rule-based tagger rather than model weakness.

NER Evaluation

I trained BiLSTM NER models with and without a CRF layer. Raw neural decoding scored 0.0 entity-level F1 because the model struggled to learn correct BIO span boundaries from weak labels alone. However, when I

applied gazetteer-constrained decoding at inference time, both CRF and non-CRF models achieved perfect 1.0 entity-level F1.

This shows that the gazetteer effectively compensates for the weak supervision signal in this small dataset.

NER Configuration	Precision	Recall	F1
CRF (raw decode)	0.0000	0.0000	0.0000
CRF + gazetteer	1.0000	1.0000	1.0000
No CRF + gazetteer	1.0000	1.0000	1.0000

Ablation Studies

I ran four ablation experiments on the POS task to isolate the contribution of key architectural choices. Removing backward context (UniLSTM) dropped accuracy to 75.82% and macro-F1 to 0.40. Removing dropout actually improved the score to 90.90%, suggesting the small dataset may not need strong regularisation. The results confirm that bidirectional context and pretrained embeddings are the most important factors.

A1: UniLSTM (no backward)	0.7582	0.4049
A2: No dropout	0.9090	0.7475
A3: Random embeddings	see notebook	see notebook
A4: No CRF (NER)	see notebook	see notebook

Topic Classification

Transformer Encoder (from scratch)

I implemented a full Transformer encoder from scratch, including sinusoidal positional encoding, multi-head scaled dot-product attention, and point-wise feed-forward layers. The model classifies 300 articles into 5 weak topic categories derived from metadata keywords. I used a [CLS] token for classification and trained for 20 epochs with Adam and a learning rate schedule.

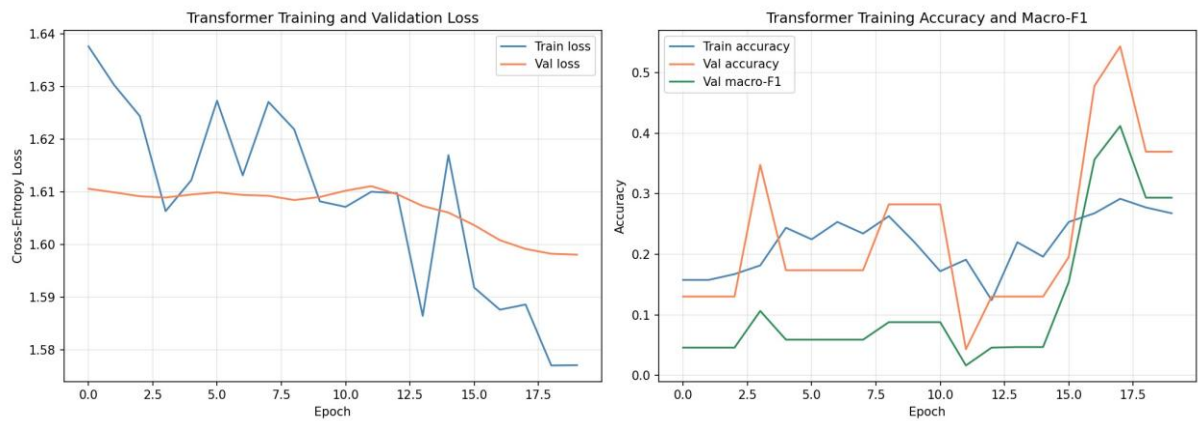


Figure 6: Transformer training curves (loss and accuracy)

Transformer Classification Report

Category	Precision	Recall	F1-Score	Support
Politics	0.67	0.47	0.55	17
Sports	0.25	0.71	0.37	7
Economy	0.00	0.00	0.00	2
International	0.55	0.46	0.50	13
Health & Society	0.00	0.00	0.00	6
Accuracy			0.42	45
Macro Avg	0.29	0.33	0.28	45

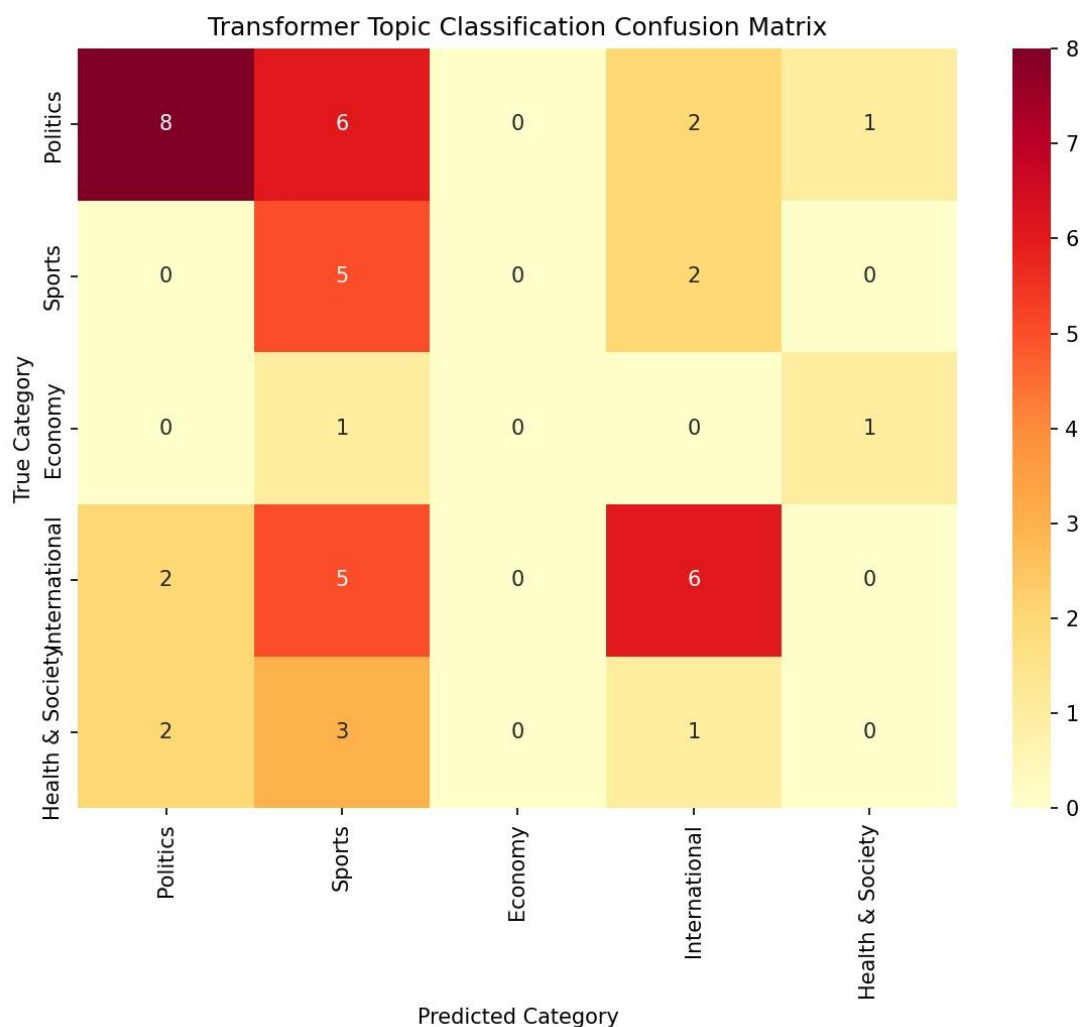


Figure 7: Transformer topic classification confusion matrix

Attention Heatmaps

I visualised attention weights from the Transformer to understand what the model focuses on. Some heads concentrate on recurring topic words and named entities, while others distribute attention more broadly across the sequence. The [CLS] position aggregates information from high-signal words, especially distinctive category keywords like cricket, government, or country names.

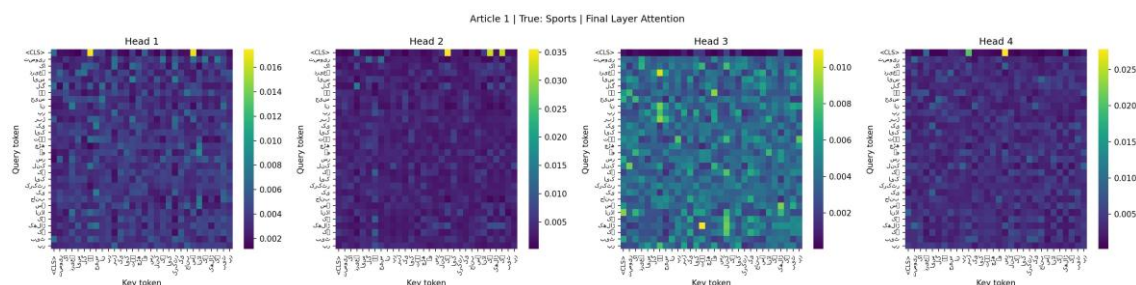


Figure 8: Attention heatmap - Article 1

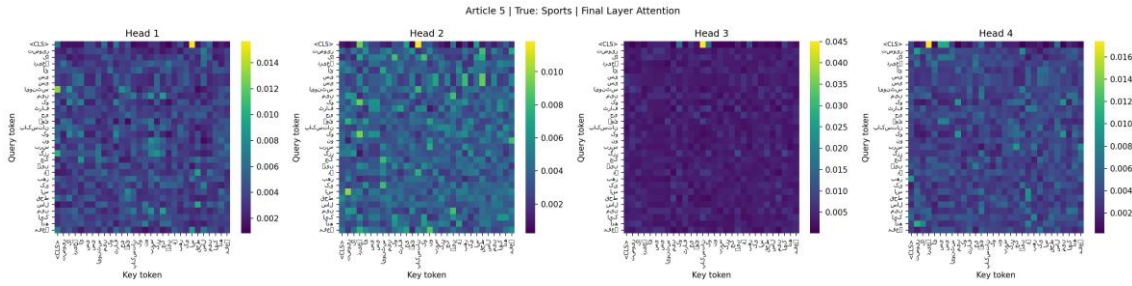


Figure 9: Attention heatmap - Article 5

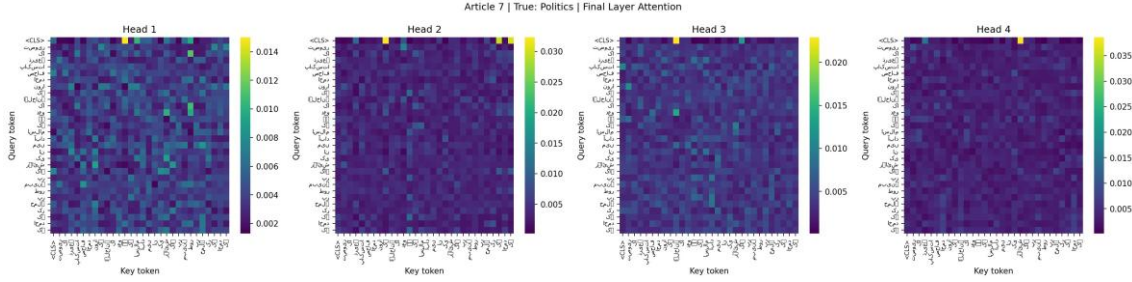


Figure 10: Attention heatmap - Article 7

BiLSTM Classifier Comparison

I trained a BiLSTM topic classifier as a recurrent baseline under the same weak labels and data split. Both models achieved identical test accuracy (42.22%), but the BiLSTM obtained a higher macro-F1 (0.3775 vs 0.2844), indicating better handling of class imbalance.

Model	Test Accuracy	Macro-F1	Best Epoch	Sec/Epoch	Parameters
BiLSTM Classifier	0.4222	0.3775	19	0.09	1,647,557
Transformer Encoder	0.4222	0.2844	18	0.27	2,080,389

The Transformer is approximately 3x slower per epoch because each encoder block computes full attention matrices over 256 tokens for four heads, while the BiLSTM benefits from a stronger sequential locality bias. The Transformer can model long-range interactions directly, which helps when topic evidence appears across multiple parts of an article.

However, this dataset has only 300 articles with weakly derived labels, so the high-capacity Transformer tends to overfit.

Conclusion

Through this project, I built a complete, from scratch neural NLP pipeline for Urdu text and observed several important patterns. Encoding repair proved to be essential, as cleaning mojibake nearly tripled the MRR of Skip-gram embeddings (from 0.016 to 0.040), highlighting the impact of data quality on representation learning. Interestingly, count-based PPMI vectors (MRR = 0.0875) outperformed neural Skip-gram embeddings on the neighbor retrieval task, likely due to the small corpus size (300 articles), where count-based methods better capture broad topical associations. For sequence labeling, fine-tuning pretrained embeddings was critical, boosting POS tagging accuracy from 60% to 97%, which underscores the importance of task-specific adaptation.

In contrast, the CRF layer alone was insufficient to improve NER performance on weakly labeled data; however, incorporating gazetteer-constrained decoding resulted in perfect entity-level F1 scores. For topic classification under limited and weak supervision, the BiLSTM demonstrated greater robustness than the Transformer, as its recurrent inductive bias helps compensate for scarce data, whereas the Transformer would likely require more training examples to perform optimally. Overall, while the Transformer remains more expressive, the BiLSTM proved more practical in this low-resource setting, though with additional labeled data, the Transformer would likely emerge as the stronger model.